Recall Radar Canada: Analyzing and Classifying Product Recall Patterns Using Canadian Safety Alert Data

Group 3

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Introduction

Product recalls are important because they protect consumers from unsafe food, health products, vehicles, cosmetics, children's products, and household items. In Canada, recall information is publicly available through the federal Recalls and Safety Alerts system, which covers product areas such as food, consumer products, health products, medical devices, cannabis, and vehicles sold in Canada (Government of Canada, n.d.-a). The database also includes recall information from Health Canada, the Canadian Food Inspection Agency, and Transport Canada (Health Canada, n.d.).

Although these notices are available online, the information is often viewed one recall at a time. This makes it difficult to understand larger patterns, such as which product categories are recalled most often, what types of safety concerns appear repeatedly, and whether recall text can be used to classify alerts more efficiently. This project proposes Recall Radar Canada, a concise analytics and machine learning project that turns public recall notices into useful trends, safety themes, and category-level insights.

Problem Statement

The main problem is that Canadian recall notices contain valuable information, but the patterns are not always easy to see without structured analysis. Consumers may want to know what types of products are most frequently recalled, while retailers and safety teams may want to monitor recurring issues such as undeclared allergens, contamination, fire hazards, choking risks, labelling problems, or product defects. A data-driven view can help organize this information and make recall monitoring easier to understand.

This project will address the problem by analyzing official Canadian recall and safety alert data, applying text analytics to identify common recall themes, building a simple classification model, and creating an interactive dashboard for practical decision-making.

Research Question and Hypotheses

The primary research question is: Can Canadian recall and safety alert data be used to identify product recall trends, classify recall notices, and highlight high-priority safety themes for better public awareness?

The null hypothesis is that recall notice text and metadata do not contain meaningful patterns that can be used to classify recall categories or identify safety themes. The alternative hypothesis is that recall notice text and metadata contain meaningful patterns that can support recall classification and theme identification with useful accuracy.

Project Objectives

1. Collect official Canadian recall and safety alert data from Government of Canada open data sources.
2. Clean and prepare the dataset for analysis by handling missing values, duplicates, date formats, and text fields.
3. Analyze recall trends by date, category, alert type, and responsible agency.
4. Use text analytics to identify common recall keywords and themes.
5. Build and evaluate machine learning models to classify recall notices into product categories.
6. Develop a transparent attention-priority score to group notices into low, medium, and high monitoring priority.
7. Create a Power BI or Streamlit dashboard that presents trends, safety themes, model results, and recommendations.

Data Source

The main dataset will be the Government of Canada Recalls and Safety Alerts dataset from the Open Government Portal. The dataset is available in structured formats such as JSON and CSV, making it suitable for data cleaning, analysis, and modelling (Government of Canada, n.d.-b). The project will use publicly available data only and will not collect personal or private consumer information.

Expected fields include recall title, publication date, product category, alert type, responsible agency, product description, reason for recall, and reference URL. The final dataset may be limited to English-language records and a selected time period to keep the project manageable.

Proposed Methodology

First, the recall data will be downloaded from the Government of Canada Open Data Portal and stored in a structured format. The data will then be cleaned by removing duplicate records, standardizing dates and categories, and preparing text fields for analysis.

Next, exploratory data analysis will be used to study recall volume over time, top product categories, agency-level patterns, seasonal trends, and common recall reasons. Text analytics will then be applied to the title and description fields using preprocessing techniques such as lowercasing, tokenization, stop-word removal, and TF-IDF vectorization.

For the modelling stage, simple and explainable machine learning models such as Logistic Regression, Naive Bayes, Decision Tree, Random Forest, and Gradient Boosting will be tested. The main modelling task will be category classification based on recall text and selected metadata. Model performance will be evaluated using accuracy, precision, recall, F1-score, and a confusion matrix.

Finally, a transparent attention-priority score will be developed using selected features such as alert type, product category, safety-related keywords, and frequency of similar themes. This score will not replace official government risk ratings. It will only be used as an analytical tool to help organize recall notices into low, medium, and high monitoring priority groups.

Expected Dashboard and Deliverables

The final dashboard will include a recall overview page, category trend analysis, safety theme explorer, attention-priority view, and model performance page. The goal is to make the results understandable for both technical and non-technical users.

Expected deliverables include a cleaned recall dataset, exploratory analysis, text analytics output, machine learning classification model, attention-priority framework, interactive dashboard, final report, poster presentation, and practical recommendations.

Expected Outcomes and Project Value

This project is expected to show that Canadian recall and safety alert data can be used to identify meaningful patterns in product safety. The project will highlight which product categories are most frequently recalled, what safety concerns appear most often, and how recall text can support classification and monitoring. The value of Recall Radar Canada is that it is practical, socially relevant, Canadian-focused, and manageable within the capstone timeline. It also demonstrates core data analytics skills, including data cleaning, exploratory analysis, natural language processing, machine learning, dashboard design, and recommendation development.

Project Timeline

Week	**Planned Work**
Week 1	Proposal approval, dataset review, and scope finalization
Week 2	Data collection, cleaning, and preparation
Week 3	Exploratory data analysis and trend analysis
Week 4	Text preprocessing and safety theme extraction

Week 5	Machine learning model development and evaluation
Week 6	Attention-priority score and dashboard design
Week 7	Dashboard refinement, recommendations, and report writing
Week 8	Final editing, poster preparation, and submission readiness

References

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Health Canada. (n.d.). Advisories, warnings and recalls - drugs and health products. Government of Canada. https://www.canada.ca/en/health-canada/services/drugs-health-products/advisories-warnings-recalls.html